

Drug safety in pregnancy

SUMMARY

Drugs can affect the fetus in various ways, with the timing of exposure during pregnancy a key factor in determining both if and how a drug will impact a developing fetus.

The exclusion of women of childbearing age from clinical trials, and the challenges in conducting large epidemiological studies, have resulted in a paucity of data on the fetal and maternal safety of drugs in pregnancy.

In some patients, the benefits of drug treatment may outweigh the potential risks to the fetus. It is important for prescribers to assess and communicate the benefits and risks in the context of the individual patient.

The Australian categorisation system for prescribing drugs in pregnancy was implemented to guide prescribers; however, it has shortcomings and lessons can be learned from the systems of other countries.

Obstetric drug information services and other evidence-based resources are available to provide guidance to healthcare professionals and consumers on the use of drugs in pregnancy and breastfeeding.

Introduction

In the 1960s, over 10,000 babies worldwide were born with severe deformities after their mothers took thalidomide to treat nausea and vomiting, a common condition of early pregnancy. Over 60 years later in November 2023, Australian Prime Minister Anthony Albanese delivered a national apology to the survivors of thalidomide and their families.¹

At the time of the thalidomide tragedy, Australia did not have a system to assess the safety of drugs before marketing. Subsequently, Australia (and other jurisdictions) implemented regulatory changes and pharmacovigilance strategies to prevent another similar tragedy. This included establishing the Therapeutic Goods Administration to regulate therapeutic goods in Australia and oversee the evaluation of drugs for safety and efficacy before and after they are marketed.

The thalidomide tragedy influenced public perception of the risks of drug exposure in pregnancy. Healthcare professionals may be hesitant to prescribe medicines to pregnant women. Some pregnant women may choose not to take necessary medicines, risking undertreatment of serious conditions such as depression,² and some may even terminate wanted pregnancies because of fears of teratogenicity.³

This article summarises the ways in which drugs can affect the fetus and how the safety of drugs in

pregnancy is assessed. The article also discusses the usefulness and limitations of the Australian categorisation system for prescribing medicines in pregnancy, and provides information on evidence-based resources for healthcare professionals and consumers on the safety of drug exposures in pregnancy and breastfeeding.

How drugs can affect the fetus

Drugs can affect the developing fetus in different ways. Teratogenic drugs interfere with fetal development and can cause congenital (structural) malformations (i.e. birth defects) following exposure during pregnancy. They can also increase the risk of other adverse pregnancy outcomes, including miscarriage, fetal growth restriction and stillbirth. The maternal use of some drugs, such as antiepileptic drugs (mainly sodium valproate), has been associated with neurodevelopmental complications (e.g. autism, intellectual disability).⁴ Other adverse effects on the fetus that may be associated with drug exposures during pregnancy are outlined in Table 1.

Drugs are not the only teratogens; infectious agents (e.g. cytomegalovirus, rubella, syphilis, toxoplasmosis), chemicals, heavy metals (e.g. lead, methylmercury), radiation, environmental factors (e.g. hyperthermia) and alcohol can also be teratogens.¹³

Gestational timing is the most important factor in determining whether a drug will affect a developing

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Table 1 Effects that drugs can have on the fetus with examples⁵

Effect on the fetus	Examples
teratogenic effects (congenital malformations)	isotretinoin, lithium, sodium valproate, warfarin
poor neonatal adaptation syndrome (generally mild and self-limiting) [NB1]	selective serotonin reuptake inhibitors, tricyclic antidepressants
neonatal abstinence syndrome (i.e. withdrawal) [NB2]	benzodiazepines, opioids
effects related to the drug's pharmacological action	angiotensin converting enzyme inhibitors, angiotensin II receptor blockers (fetal renal impairment with exposure after first trimester) ^{6,7} azathioprine (transient neonatal bone marrow suppression) beta blockers (apnoea and bradycardia) nonsteroidal anti-inflammatory drugs (premature closure of the ductus arteriosus [NB3], fetal renal impairment with exposure in later pregnancy) ^{8,9} warfarin (fetal haemorrhage)

NB1: Poor neonatal adaptation syndrome occurs in one-third of newborns exposed to selective serotonin reuptake inhibitors in utero. Symptoms can include poor muscle tone, tremor, jitteriness, sleep disturbance and irritability; however, symptoms are usually mild and transient, and typically resolve without medical intervention.¹⁰

NB2: Neonatal abstinence syndrome is caused by a sudden discontinuation of fetal exposure to a drug after delivery that was used by the mother during pregnancy.^{11,12}

NB3: The ductus arteriosus is a blood vessel that connects the fetal pulmonary artery to the aorta which closes shortly after birth.⁸

fetus, and the nature of the effect.¹⁴ Other factors that may influence fetal drug effects include:¹⁵

- dose (all drugs have a no-observed-adverse-effect level)
- route of exposure (topical preparations generally result in lower systemic maternal and fetal exposure than oral or parenteral preparations)
- pharmacogenomics of the mother and fetus
- how much drug is transferred across the placenta (influenced by factors such as drug molecular weight and ionisation).

Embryology and scientific publications usually describe gestational timing as the time since conception, while clinical practice dates a pregnancy from the last menstrual period (LMP). This difference can cause confusion. The clinical description of pregnancy trimesters does not reflect important biological milestones, which can result in misleading advice around drug exposure. Therefore, fetal drug effects are best understood in relation to the periods of fetal development: the all-or-none period, the embryonic period, and the fetal period.

All-or-none period

The first 2 weeks after conception (2 to 4 weeks post-LMP) is often called the 'all-or-none' period because the exposure either results in loss of the pregnancy or no effect. It is generally believed that drug exposures during this time do not cause malformations as the conceptus is a mass of dividing stem cells that have minimal contact with the maternal circulation and have not yet differentiated into structures and organs.^{14,16}

Embryonic period

The embryonic period or period of organogenesis (between 4 and 11 weeks post-LMP) is the most critical period of fetal development,¹⁴ during which drug exposures have the greatest likelihood of causing structural birth defects, such as neural tube defects, cardiac defects and orofacial clefting, as well as miscarriage.

Most teratogenic drugs exert their teratogenic effects during a relatively small window of time. For example, thalidomide caused limb defects following exposure between 34 and 49 days post-LMP.¹⁷

Fetal period

During the fetal period (from 11 weeks post-LMP until birth), drug exposures do not cause malformations but may cause disruption of normally formed organs by several mechanisms including hypoxia, vasoconstriction, hypotension and oligohydramnios (decreased amniotic fluid).¹⁴ Generally, these effects are caused by the drug's recognised pharmacological action (see examples in Table 1).

Assessment of drug safety in pregnancy

It is difficult to categorically prove whether or not a drug is teratogenic as no single approach can definitively establish the safety or otherwise of drugs in human pregnancy.¹⁸ Following the thalidomide tragedy, it was required that preclinical (animal) studies included at least 2 different species, of which one had to be primate. While data from animal studies cannot always be extrapolated to humans, development of anomalies in several animal species heightens the suspicion that a drug is a human teratogen.

In 'the shadow of thalidomide', the US Food and Drug Administration (FDA) issued guidelines in 1977 that excluded women of childbearing potential (except those with life-threatening conditions) from the early phases of most clinical trials, because of the potential for fetal harm.¹⁹ This long-standing exclusion has resulted in major gaps in evidence about the safety

of drugs for pregnant and breastfeeding women. Consequently, most of this information is obtained after drugs have been marketed.²⁰

Most teratogenic drugs have been identified through single case reports or small case series.²¹ Voluntary or spontaneous reporting, including drug company registries and postmarket surveillance, can contribute to the assessment of drug safety in pregnancy but is subject to bias and incomplete reporting and follow-up. Studies linking data from birth defect and prescribing registries have been used to identify potential teratogenic signals but also have significant methodological issues.²²

An ideal approach to determine the safety of drugs in pregnancy is through large postmarket epidemiological studies; however, these can be challenging to implement. Postmarket prospective cohort studies compare pregnancy outcomes of women exposed to the drug in question with unexposed controls, and ideally include a disease-matched control group so that effects of the underlying condition can be considered. However, such studies are expensive, and it can take a long time to obtain cases (particularly of new or rare exposures) and to detect whether there are signals of concern. Retrospective case-control studies incorporate larger numbers and are more cost-effective and efficient, allowing evaluation of specific, and potentially rare, birth defects. However, because of their retrospective nature, they are subject to recall bias, and data on adverse outcomes other than malformations are lacking.²³

The Australian categorisation system for prescribing medicines in pregnancy

As a direct consequence of the thalidomide tragedy, the Australian Drug Evaluation Committee (ADEC) was established in 1963 to advise the Australian Government on the safety of new drugs and to monitor and evaluate potential adverse effects of already available drugs. The Australian categories for prescribing medicines in pregnancy (A, B1, B2, B3, C, D, X) were established by the ADEC and are still in use today, significantly influencing prescribing and how risks of drug exposures in pregnancy are perceived by healthcare professionals and consumers.^{24,25}

While providing guidance for prescribers, the A to X categorisation is overly simplistic and may misleadingly imply there is a hierarchy of risk. For example, a drug labelled category B is not necessarily safer than a drug labelled category C, which can cause anxiety and confusion for prescribers and consumers. Where there are no or limited human data, the categories are assigned on the basis of

data from animal studies and are not updated regularly, even when there are new (and potentially reassuring) human data available. Drugs within the same category do not always carry a similar level of risk, and the categories do not consider the timing of pregnancy. Furthermore, they are not relevant for breastfeeding.^{26,27}

Because of these inherent problems, the FDA abandoned their A to X categorisation in 2015 in favour of a descriptive labelling system. The Pregnancy and Lactation Labelling Rule provides a more consistent way of presenting relevant reproductive safety information about drugs, enabling prescribers to better understand and communicate the risks and benefits of drug treatment to their patients.²⁸

Sources of information on the safety of drug exposures in pregnancy

Drug information services and evidence-based resources are available in Australia and internationally to provide guidance to healthcare professionals and consumers on the safety of drug exposures during pregnancy and breastfeeding (Table 2).

Obstetric drug information services provide free telephone advice and are available in most states and territories in Australia. There are also teratogen information services in North America and Europe. These services provide free information to consumers and healthcare professionals on drug exposures during pregnancy and breastfeeding.²⁹ However, they cannot always provide adequate information or reassurance as there are often limited data on the use of drugs in pregnancy, especially for new drugs. Teratogen information services also undertake research on drugs in pregnancy.³⁰

Online reproductive databases provide updated and well-referenced summaries about exposures during pregnancy (e.g. Reprotox, Teratogen Information System) and breastfeeding (e.g. Lactmed).

Approach to the pregnant patient

When deciding whether to initiate, continue or discontinue a drug in a pregnant woman, it is important to conduct an individualised assessment of the benefits and risks to the mother and fetus. Factors that should be considered include the severity of the mother's medical condition, risks of the uncontrolled, untreated or inadequately treated maternal condition, risk and severity of adverse drug effects on the fetus, gestational timing, and patient preferences.³¹ For example, in women already taking an antidepressant (e.g. selective serotonin reuptake inhibitor), the benefits of treating the depression often outweigh the potential risk of poor neonatal adaptation syndrome. Uncontrolled depression has been associated with

Table 2 Sources of information for healthcare professionals and consumers on the safety of drug exposures during pregnancy and breastfeeding

Source of information	Example	Cost
Obstetric drug information services (Australia) [NB1]	Australian Capital Territory – Australian Capital Territory Drug Information Service, The Canberra Hospital	Free
	New South Wales – MotherSafe, Royal Hospital for Women	Free
	Northern Territory – Northern Territory Drug Information Centre, Royal Darwin Hospital	Free
	Queensland – Queensland Medicines Advice and Information Service, Royal Brisbane and Women's Hospital	Free
	South Australia – Medicines and Drug Information Centre, Women's and Children's Hospital	Free
	Victoria – Drug Information Services, The Royal Women's Hospital; Drug Information Centre, Monash Medical Centre	Free
	Western Australia – Obstetric Medicines Information Service, King Edward Memorial Hospital	Free
	Tasmania – no service currently available	Free
Teratogen information services (outside of Australia)	MotherToBaby (United States)	Free
	European Network of Teratology Information Services (ENTIS)	Free
Online databases	Teratogen Information System (TERIS)	Subscription-based
	Reprotox	Subscription-based
	Reprotext	Subscription-based
	Shepard's Catalog of Teratogenic Agents	Physical or digital product available for purchase
	Best use of medicines in pregnancy (Bumps)	Free
	Lactmed (information about the use of drugs in breastfeeding)	Free
Reference guides	Australian Medicines Handbook (AMH)	Physical or digital product available for purchase
	The Women's Pregnancy and Breastfeeding Medicines Guide	Subscription-based
	Briggs Drugs in Pregnancy and Lactation	Physical or digital product available for purchase

NB1: Further information about each Australian state or territory's obstetric drug information service can be found on the [Therapeutic Goods Administration](#) website.

adverse perinatal outcomes including preterm birth, low birth weight and potential impacts on child neurodevelopment.³²⁻³⁴ Additionally, women with severe depression are at a higher risk of relapse during pregnancy if their antidepressant is discontinued.³⁵

Examples of other conditions where drug treatment is important during pregnancy include epilepsy, diabetes, and autoimmune conditions such as inflammatory bowel disease. Patient counselling, and

advice from an obstetric drug information service, can help address patient concerns and contextualise the benefits and risks for the individual patient.³⁶

Conclusion

Drugs can have different effects on the fetus during pregnancy, which are dependent on a range of factors including the timing of exposure during fetal development. It remains challenging to obtain robust

evidence for drug safety in pregnancy as women of childbearing age are generally under-represented in clinical trials. It is essential to weigh the benefits against the potential risks of drug treatment in the context of the individual patient, including risks associated with the undertreated maternal medical condition, and risks to the pregnancy and fetus. While the Australian categorisation system for prescribing drugs in pregnancy can be helpful, there are important limitations. Obstetric drug information services and other evidence-based resources are available to provide guidance to healthcare

professionals and consumers regarding drug exposures in pregnancy and breastfeeding. ◀

Conflicts of interest: Debra Kennedy is the Director of MotherSafe, a statewide teratogen information service based at the Royal Hospital for Women, which provides telephone advice and counselling to women and healthcare providers on drug and nondrug exposures during pregnancy and breastfeeding.

Ronald Batagol has published several reference guides on the safety of drugs used in pregnancy.

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